

UEI742: SMART SENSOR NETWORKS

L	T	P	Cr
3	0	0	3.0

Course Objectives: The objective of the course is to provide basic understanding of sensor networks, standards and protocols, application and latest trends.

Introduction: Basics of wireless sensor network, Sensor network architecture, Open Systems Interconnection model, components of sensor network nodes, Advantage of Sensor Networks, Applications of Sensor Networks, Reconfigurable Sensor Networks, Military application, Habitat monitoring, Wireless sensor network as embedded system, Tiered architecture in sensor network, Routing and addressing in tiered architecture, Energy efficient design of Wireless sensor nodes.

Fundamentals of Medium Access Control Protocols, Classification of MAC Protocols, S-MAC Protocol, B-MAC protocol, A survey of MAC protocol for sensor network, IEEE 802.xx standard, IEEE 802.15.4 standard and Zig Bee, IEEE 1451 Family of Smart Transducer Interface Standards

Taxonomy of routing techniques in sensor networks: Routing Protocols/ strategies in WSN, Sensor Protocols for information vis negotiation, Low energy adaptive clustering, hierarchy (LEACH), Data dissemination and gathering, Power efficient gathering sensor information system, Reliable Transport in Sensor Networks, Dissemination Protocols for large sensor networks.

Survey of Latest Protocol: Features of recent networks such as WAN – GPRS, WAN - 3G, WAN-4G, WAN, Wi Fi/WLAN 802.11xx, Wi Fi/WLAN – AP, Wi Fi/WLAN – STA, Wi Fi/WLAN - P2P, Wi Fi/WLAN – WPS, Wi Fi/WLAN - WPA personal, Wi Fi/WLAN - WPA enterprise, Bluetooth Classic Vx.x, Bluetooth Low Energy Vx.x, Dual mode Bluetooth Vx.x, 6LoWPAN, Brillo, Echo, Homekit, iAP1, iAP2, LoRaWAN V1.0, Neul, NFC, SigFox, Thread, ZigBee 3.0, Z-wave.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Exhibit the knowledge of the concept of sensor network and its application
2. Classify the communication Protocols in sensor networks
3. Apply the concept of routing techniques in sensor networks
4. Implement the concept of sensor networks in the current advance applications like WiFi, Bluetooth, Homekit, ZigBee and Z-wave etc.

Text Books:

1. H. Karl and A. Willig, *Protocols and Architectures for Wireless Sensor Networks*.
2. John Wiley & Sons, June 2005.
3. K. Sohrawy, D.Minoli, and T. Znati. *Wireless Sensor Networks: Technology, Protocols, and Applications*. John Wiley & Sons, March 2007.

4. C. S. Raghavendra, K. M. Sivalingam, and T. Znati, *Wireless Sensor Networks*. Springer Verlag, Sep. 2006.
5. B. Krishnamachari, *Networking Wireless Sensors*.. Cambridge University Press, Dec. 2005.
6. D.P. Agrawal, *Embedded Sensor Systems*, Springer, 2017.

Reference Books:

1. Mohammad Ilyas and Imad Mahgoub, *Handbook of Sensor Networks: Compact Wireless and Wired Sensing systems*, CRC PRESS, 2004

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30